

Vaishnavi Balay

Full Stack Developer

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SUMMARY

Full Stack Developer with Over 3 years of experience leading projects through the full software development lifecycle (SDLC), with expertise in designing, developing, implementing, and deploying solutions that ensure seamless integration between front-end and back-end systems for successful project delivery. Skilled in the Software Development Life Cycle (SDLC), Agile/Scrum methodologies, and programming languages, including Java, Python, JavaScript, and SQL for developing robust applications. Proficient in developing end-to-end applications using Spring Boot, Spring REST, Spring Cloud, Spring Security, Spring Data JPA, Spring MVC, Hibernate, and RESTful APIs, integrated with dynamic user interfaces built using React.js. Experienced in testing, version control, and project management tools, including Postman, JIRA, JUnit, React Testing Library, Mockito, Git, and PostgreSQL for building, testing, and maintaining robust applications.

EDUCATION

Masters in Computer Science

University of Texas at Arlington, TX

May 2024

Bachelor of Technology in Computer Science

Jawaharlal Nehru Technological University, Hyderabad, India

May 2021

TECHNICAL SKILLS

Languages: C, C++, Java, JavaScript, SQL, TypeScript

Frameworks/Libraries: Hibernate, JSF, React.js, Next.js, Angular, Node.JS, GraphQL, Mockito

Spring Frameworks: Spring Boot, Spring MVC, Spring REST, Spring Data JPA, Spring Batch, Spring Security

J2EE Technologies: Servlets, JSP, JSTL, JavaBeans, JDBC

Web Technologies: HTML5, Tailwind CSS, JavaScript, Bootstrap5, jQuery, Ajax, JSON, XML, Redux

Database: Oracle, MySQL, PostgreSQL, MongoDB

Cloud: AWS (EC2, SQS, SNS, Elastic Beanstalk, Lambda, Code Deploy, CloudWatch)

CI/CD and DevOps Tools: Maven, Gradle, Git, GitHub, Jenkins, Postman, Junit, Docker, Kubernetes, Kafka

Methodologies: SDLC, Agile/Scrum, Waterfall, TDD

PROFESSIONAL EXPERIENCE

Freddie Mac, USA

July 2024 – Present

Full Stack Developer

- Applied Spring MVC applications with relational and NoSQL databases using Spring Data JPA and Hibernate, optimizing ORM mapping, lazy/eager loading strategies, and query performance tuning to enhance database query efficiency by 40%.
- Boosted centralized authentication and authorization service using Spring Security and OAuth2 Authorization Server, improving system-wide security consistency and reducing integration efforts by 50%.
- Developed RESTful APIs using Spring Boot to enable seamless communication between distributed systems, enhancing system integration, request handling, and service interoperability improvement in application performance 35% increase in scalability, and enhancement in the reliability of backend services.
- Configured real-time alerts and dashboards in CloudWatch to track application performance, API response times, and system health, reducing incident resolution time by 20% through proactive monitoring.
- Streamlined fine-grained authorization and authentication mechanisms for GraphQL APIs using Spring Security and JWT, ensuring secure access control and data protection authentication workflows, reducing unauthorized access incidents by 30%.
- Performed JUnit with CI/CD pipelines using Jenkins, GitLab CI/CD, and GitHub Actions, automating test execution and reducing deployment failures by 50% through continuous testing and early defect detection.

DBS Bank, India

Dec 2021 – Jul 2022

Full Stack Developer

- Enhanced front-end development using HTML5, CSS3, JavaScript, jQuery, Bootstrap, and AngularJS to build responsive UIs, collaborating with designers to translate UI/UX concepts into functional web pages.
- Built RESTful APIs using Spring MVC and Spring WebFlux for efficient data communication between microservices and the front end, achieving a 30% increase in API request throughput.
- Designed and maintained Spring MVC-based applications, improving request handling efficiency and reducing API latency by refactoring controllers, services, and repository layers.
- Collaborated with cross-functional teams to optimize applications and IT operations within the bank, improving system performance and reliability.
- Implemented efficient CRUD operations for entities, utilizing database normalization and indexing, resulting in a robust and scalable data layer for applications, contributing to a 20% improvement in application performance.
- Accelerated ECS integration with Elastic Load Balancing, CloudWatch, and CloudTrail to create a highly scalable and resilient web application architecture, achieving a 99.99% uptime and reducing operational costs by 20%.

Accenture, India

Aug 2021 – Dec 2021

Application Development Associate

- Optimized dynamic, high-performance front-end applications using React.js, improving user engagement and reducing page load time by 30% through efficient component rendering and lazy loading techniques.
- Engineered scalable microservices and enterprise applications using Spring Boot, leveraging a modular architecture and embedded servers to streamline deployment processes, resulting in a 40% reduction in deployment time and improved system scalability.
- Built OAuth 2.0 authentication using Spring Security to secure external application access, enhancing API security and user authentication through token validation and role-based access control (RBAC), contributing to a 30% improvement in access management efficiency.
- Increased request validation, transformation, and mapping templates using AWS API Gateway, reducing manual backend processing time by 30% and ensuring standardized API request/response formats.
- Collaborated with Product, Test, and DevOps teams to ensure seamless integration, leveraging Agile methodologies to deliver high-quality software solutions.
- Enhanced test case execution by integrating Mockito with JUnit, optimizing dependency mocking, and automating test setups, resulting in a 40% reduction in test setup time and significantly improving overall testing efficiency and maintainability.

Bytes Soft Solution, India

Jan 2020 – Jul 2021

Jr. Java Full Stack Developer

- Leveraged RxJS Observables to handle asynchronous data flows in a React application, improving user experience by enhancing responsiveness by 25%.
- Created detailed technical documentation in JIRA Confluence, including architecture diagrams, API specifications, and development guidelines, improving knowledge sharing and onboarding efficiency.
- Crafted and maintained well-documented GraphQL APIs with clear and concise schema definitions, improving the developer experience and reducing integration time for frontend and applications.
- Implemented comprehensive unit and integration tests using Java Spring Boot (JUnit), automating backend API calls to enhance product reliability and reduce testing redundancy by 40%.
- Improved message delivery reliability by 50% through robust message queuing and routing mechanisms provided by RabbitMQ in the microservices communication layer.